



# Alna Eldhose *AI Systems & Operations Engineer*

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📍 Cottbus, Germany

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## 📄 SUMMARY

M.Sc. Artificial Intelligence student at BTU Cottbus-Senftenberg with enterprise IT experience at Raisin SE. Combines deep learning model development with strong foundational systems engineering, identity management, and infrastructure administration. Experienced in optimizing edge AI models, managing enterprise SaaS tooling, and automating operational workflows. Seeking full-time roles bridging machine learning engineering, IT systems, and automation.

## 🧠 SKILLS

**Programming Languages** (Python, C++, Java, SQL, Bash/Shell)

**Core Computer Science** (Object-Oriented Programming (OOP), Data Structures & Algorithms, RDBMS (MySQL))

**Machine Learning** (Continual Learning, PEFT (LoRA, TT-LoRA), Vision Transformers (ViT), Semi-Supervised Learning, Audio Classification, Model Robustness)

**Frameworks & Libraries** (PyTorch, TensorFlow, Scikit-learn, Librosa)

**Cloud & Systems** (Linux/Unix, SLURM, Git, GitLab, Automox, Snipe-IT)

**Enterprise IT & Identity** (Okta (IAM/SSO), Google Workspace, DocuSign, Jamf, Microsoft Intune, Jira Service Management, Confluence)

## 🌐 LANGUAGES

**English:** Proficient

**German:** Basic

**Malayalam:** Native

## 📁 PROFESSIONAL EXPERIENCE

### **Transformer-Based Semi-Supervised Incremental Learning for Edge Devices via Tensor-Train LoRA Adapters**, BTU Cottbus-Senftenberg

08/2026 – Present

*Master's Thesis*

- Developing a continual learning pipeline using Tensor-Train LoRA (TT-LoRA) adapters on a frozen MobileViT backbone to enable lightweight, low-footprint model updates for edge deployment.
- Scaling semi-supervised representation learning on large unlabeled image datasets using adaptive thresholding and entropy filtering.
- Formulating Fisher-guided layer sensitivity adaptation to improve model robustness under hardware-level bit-flip faults.

### **Working Student – Enterprise IT Support Engineer**, Raisin SE

02/2024 – Present | Berlin, Germany

- Administer identity management, user lifecycles, and access governance across Okta, Google Workspace, and Jira Service Management for enterprise stakeholders.
- Manage endpoint fleets and device compliance across macOS and Windows using Jamf and Microsoft Intune, maintaining accurate hardware assets in Snipe-IT.
- Resolve and escalate cross-functional infrastructure tickets, maintaining network uptime, workstation environments, and internal tooling.
- Author technical SOPs and internal system documentation to standardize provisioning workflows and improve knowledge transfer.

## 🎓 EDUCATION

### **M.Sc. Artificial Intelligence**, BTU Cottbus-Senftenberg

10/2022 – Present | Cottbus, Germany

### **B.Sc. Computer Application**, Nirmala College

07/2018 – 08/2021 | Muvattupuzha, India

## 📁 PROJECTS

### **Musical Instrument & Pitch Recognition via CNNs**

10/2024 – 02/2025

BTU Cottbus-Senftenberg

- Built an end-to-end audio classification pipeline using Convolutional Neural Networks to classify 14 instruments and detect musical pitch.
- Extracted time-frequency acoustic representations by transforming raw audio signals into Mel-spectrograms using Librosa.
- Engineered multi-layer convolutional and max-pooling feature extractors connected to dense classification heads targeting an 80%+ benchmark.